Supratik Bhattacharya

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Summary

Research Engineer II at Microsoft focused on LLM evaluation, model post-training, and enterprise security retrieval. Experience designing experiments, building synthetic benchmarks, and analyzing model failures, with publications at Interspeech, COLING, and ICMI.

Technical Skills

Machine Learning: Python, PyTorch, C++, multimodal learning, representation learning
LLMs: SFT, LoRA/PEFT, teacher–student distillation, quantization, LLM evaluation, judge models, tool use, context compression, guardrail evaluation
Systems / Retrieval: Linux, Docker, Kubernetes, Azure ML, vLLM, CPU inference, RAG, embeddings, semantic retrieval, multi-GPU training

Industry Experience

Research Engineer II, Microsoft – Bengaluru, India Apr 2024 – Present

- Evaluated small open-weight LLM judges against larger-model baselines across multilingual objective and subjective criteria, identifying systematic evaluation failures and improving accuracy from **79.6%** to **96.2%** through rubric redesign and failure-mode analysis.
- Built an adversarial retrieval benchmark using LLM-generated synthetic data, semantic rewrites, and character obfuscation; evaluated robustness with Recall@K and MRR over a curated 20K-sample phishing dataset.
- Designed and analyzed experiments for LLM guardrails, evaluating failure modes and behavioral regressions across safety- and reliability-focused test cases.
- Developed post-training pipelines for Google's **Gemma** open-weight models using supervised fine-tuning (LoRA), teacher–student distillation, and quantization for judge-model and tool-use tasks on Azure ML.
- Compared embedding-fusion and graph-aware ranking strategies for semantic retrieval in enterprise security, selecting PCA-based fusion after controlled experiments showed attention-based fusion overfit at 20K samples.
- Collaborated with Microsoft Research to evaluate **GraphRAG** and **LazyGraphRAG** for an enterprise security retrieval service.
- Built large-scale embedding pipelines supporting **1M+ embeddings/day** for enterprise retrieval workloads.
- Investigated context-management strategies for live voice agents with 75+ tools, evaluating prompt compression, tool-schema restructuring, and retrieval-based selection of tool descriptions.
- Improved internal benchmark tool-call success from **60%** to **89%** and eliminated hallucinated tool-argument identifiers (**48** $\rightarrow$ **0** per session) through system-prompt compression, tool-schema restructuring, and retrieval-based context selection.
- Prototyped and evaluated LLM inference optimizations including **KV-cache management, vLLM serving, speculative decoding, and CPU inference**, with emphasis on latency, throughput, and deployment efficiency.

Software Engineer, Cisco Systems – Bengaluru, India Aug 2022 – Apr 2024

- **Prototyped** an LLM-based network log summarization tool for incident triage and **refactored** legacy C/C++ authentication components in IOS XE, reducing end-to-end latency by $\sim$**40%** under production workloads.

Research Experience

Research Collaborator, University of Toronto – Matter Lab Jul 2021 – December 2021

- **Developed** a photorealistic 3D image-generation pipeline to create synthetic datasets for a self-driving chemistry lab; used the generated data to evaluate semantic-segmentation models for recognising labware and reaction states.

Research Intern, Samsung R&D Institute – Bengaluru, India May 2021 – Jul 2021

- **Developed** segmental spectrum warping and formant-energy perturbation to augment adult speech for children's-speech ASR; multi-GPU training. Published at Interspeech 2022.

Research Intern, University of Hamburg May 2020 – Jul 2020

- **Built** a multimodal (audio + video) attention-based PyTorch classifier for on-screen vs. off-screen speech to selectively apply lip-sync constraints in machine dubbing; also contributed to a study of synchrony-constraint impact on NMT. Published at **COLING 2020** and **ICMI 2020**.

Selected Publications

V. P. Singh, H. Sailor, **S. Bhattacharya**, A. Pandey. *Spectral Modification Based Data Augmentation for Improving End-to-End ASR for Children's Speech.* **Interspeech 2022**.
A. Karakanta, **S. Bhattacharya**, S. Nayak, T. Baumann, M. Negri, M. Turchi. *The Two Shades of Dubbing in Neural Machine Translation.* **COLING 2020**.
S. Nayak, T. Baumann, **S. Bhattacharya**, A. Karakanta, M. Negri, M. Turchi. *See Me Speaking? Differentiating on Whether Words are Spoken On Screen or Off to Optimize Machine Dubbing.* **ICMI 2020 (DVU Workshop)**.

Education

M.Sc. Chemistry & B.E. Computer Science (Dual Degree) 2017 – 2022
Birla Institute of Technology and Science – Pilani, India *CGPA: 8.93 / 10.00*